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ARTICLE



The association between White matter microstructure alterations detected by Diffusional kurtosis imaging in Neural circuit and post-stroke depression

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ABSTRACT

Aim: In order to study the mechanism of post-stroke depression (PSD), we used diffusion kurtosis imaging (DKI) to describe the changes in white matter (WM) microstructure in PSD patients, to investigate the association between WM damage in limbic-cortical-striatal-pallidal-thalamic (LCSPT) circuit and PSD, and the utility of DKI in the diagnosis of PSD.

Methods: Fifty-eight participants were divided into different groups: control group (n=20), stroke patients without depression (Non-PSD, n=21) and PSD group (n=17). All were taken DKI scans. The WM of bilateral superior frontal gyrus, middle frontal gyrus, inferior frontal gyrus, temporal lobe, parietal lobe, occipital lobe, the anterior and posterior limb of internal capsule, the genu and splenium of corpus callosum were selected as the regions of interest (ROI) and selected mean kurtosis (MK), radial kurtosis (RK), axial kurtosis (AK) as the DKI parameters.

Results: Compared with control and Non-PSD, MK of PSD group in bilateral superior frontal gyrus, middle frontal gyrus, temporal lobe and the genu of corpus callosum were decreased significantly, as well as the RK in left superior frontal gyrus, bilateral middle frontal gyrus and temporal lobe. But there was no significant difference in AK. Besides, the decrease in MK and RK in frontal and temporal lobe was negatively associated with the severity of PSD.

Conclusion: Our research indicated that the damage to WM microstructure in the frontal lobe, temporal lobe and the genu of corpus callosum may be related to PSD. DKI explores the microstructural changes of WM in PSD patients and may be an auxiliary diagnostic method for PSD.

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1. Introduction

Post-stroke depression (PSD) is a common emotional complication of stroke, which is characterized by persistent depression, anhedonia, slow thinking and action, less speech, fatigue and sleep disorders [1]. Severe patients may be accompanied by suicidal ideas and behaviors, which seriously affect the prognosis of stroke patients and increase the risk of death of stroke patients [2]. About one-third of stroke patients were complicated with PSD [8]. Based on the high morbidity of PSD, it is essential to clarify the pathogenesis of PSD in order to achieve early treatment and improve the prognosis of stroke patients.

At present, the pathogenesis of PSD is still unclear. The limbic-cortical-striatal-pallidal-thalamic (LCSPT) circuit is considered to be closely associated with emotional regulation [9]. LCSPT circuit was proposed in 1972, originates from the dorsal lateral prefrontal cortex (DLPFC), is projected to the neostriatum and thalamus, then back to the cortex [10], which mainly includes limbic system, cortex, striatum, globus pallidus and thalamus. These brain structures are closely

connected with each other, and control various activities of the body. Sheline YI first discovered that there were structural changes in brain of depressive patients, which are mainly located in the LCSPT circuit [9]. Hippocampal, striatum, thalamus and deep white matter of frontal lobe are vital components of this circuit, which play an important role in regulating emotion. It has been reported that the metabolic abnormality of striatum, hippocampus and thalamus in LCSPT circuit of depressive patients may be due to synaptic dysfunction [11]. Guo H et al. reported that the volume of prefrontal cortex, anterior cingulate gyrus, basal ganglia, thalamus, hippocampus and amygdala was decreased in major depression disorder (MDD) patients [12]. Terroni L et al. demonstrated that PSD is associated with the destruction of left LCSPT circuit [13], further supporting that white matter damage under the LCSPT circuit may increase the risk of depression. The normal function of LCSPT circuit requires the integrity of the white matter fibers of the brain. When the white matter integrity is destroyed, the LCSPT circuit is impaired and results in emotional dysregulation such as depression [14].

Therefore, it is important to monitor the integrity of white matter fiber in LCSPT circuit.

At present, the main measure to study the integrity of white fiber is diffusion tensor imaging (DTI), a new magnetic resonance imaging (MRI) modality, can be used to assess the microstructure of white matter fibers in vivo by measuring the intensity and direction of the water molecules' diffusion [15]. DTI is suitable for the diffusion of water molecules in tissue conforms to the Gaussian distribution. However, due to the limitation of diffusion barriers such as cell membranes and organelles in human body, the diffusion of water molecules often deviate from the Gaussian distribution [16], which be called Non-Gaussian distribution [17]. A novel diffusion modality, diffusion kurtosis imaging (DKI) was proposed by Jensen et al. to overcome the limitations of DTI [18], to quantify the dispersion deviation from the Gaussian distribution and provide a measure of the degree of diffusion hindrance or restriction[5]). Since the deviation from Gaussian dispersion is governed by the complexity of the tissue within which the water is diffusing, this diffusional kurtosis can be regarded as a measure of a tissue's complexity of structure. Compared with DTI, DKI can supply new information to better characterize both normal and pathological brain tissue [16] and is more sensitive and specific. Li HQ et al. evaluated the sensitivities of kurtosis parameters derived from DKI in detecting microstructural changes and associated pathology, they demonstrated that the kurtosis parameters such as mean kurtosis (MK), radial kurtosis (RK) could provide complementary information for revealing brain microstructural damage [19]. Falangola MF et al. pointed out that non-Gaussian diffusion MRI may be beneficial in the assessment of microstructural tissue damage at the early stage of mild cognitive impairment [20], illustrated that DKI can be used to evaluate disease progression and prognosis [21].

Based on the above considerations, our study will use DKI to detect the change of white matter microstructure in LCSPT circuit of PSD patients, to prove the following hypothesis: Damage of white matter microstructure occurring in the LCSPT circuit will increase the risk of PSD.

2. Materials and methods

2.1. Participants

The study was approved by local institutional review boards, and written informed consents were obtained from all participants. From May 2017 to December 2017, Thirty-eight patients with acute cerebral infarction who were hospitalized at the Second Hospital of Shanxi Medical University were enrolled, following the Guidelines for the Diagnosis and

Treatment of acute ischemic stroke in China 2016; All subjects were confirmed by cranial CT or MRI examinations and were first-episode patients with onset time less than 7 days, and have the ability to complete questionnaires. Twenty healthy controls were selected. All subjects are right-handed.

2.3. Exclusion criteria

Severe physical diseases that endanger life or hinder recovery after stroke;

Communication difficulties caused by language or articulation disorders;

Any neuropsychiatric diseases, such as Alzheimer's disease, Parkinson's disease, brain tumors, epilepsy, psychosis and alcohol or substance abuse;

Somatic diseases affecting motion before stroke

2.2. Groups

Demographic data and stroke characteristics were collected at the time of admission. Two weeks after stroke onset, Hamilton Depression Scale-24 (HAMD-24) was used by two well-trained neurologists to evaluate the depression status of all subjects. The specific grouping is as follows:

Control:subjects without stroke or depression

Non-PSD: Stroke patients with HAMD score <8

PSD: met the diagnostic criteria of PSD and HAMD score ≥ 8

2.3. KI scan

MRI scans were performed using a GEMR750W 3.0 T magnetic resonance scanner, including T1 weighted images, T2 weighted images, DWI sequences, and DKI sequences. The DKI scan uses Axial scanning parameters:repetition time (TR) 6500 ms, echo time (TE) 95.80 ms, FOV 24 cm, matrix 128×128 , number of layers 25, layer thickness 3.6 mm, layer Pitch 0.4 mm, b value (0, 1250, 2500)s/mm², 25 diffusive magnetically sensitive directions, scan time 11'10". Data were analyzed by using function tool 9.4.05 post-processing software of GE AW 4.6 post-processing work station. The white matter of bilateral superior frontal gyrus, middle frontal gyrus, inferior frontal gyrus, temporal lobe, parietal lobe, occipital lobe, the anterior and posterior limb of internal capsule, the genu and splenium of corpus callosum were selected as the regions of interest (ROI)(Figure 1.), and selected the mean kurtosis (MK), radial kurtosis (RK), axial kurtosis (AK) as the DKI parameters. Each ROI was measured three times and average the results. We try to avoid the infarct zone, brain gray matter, cerebral sulcus, and vascular space when selecting the ROI, and try to satisfy all patients with the same level, bilateral symmetry, and the same area. All imaging technicians

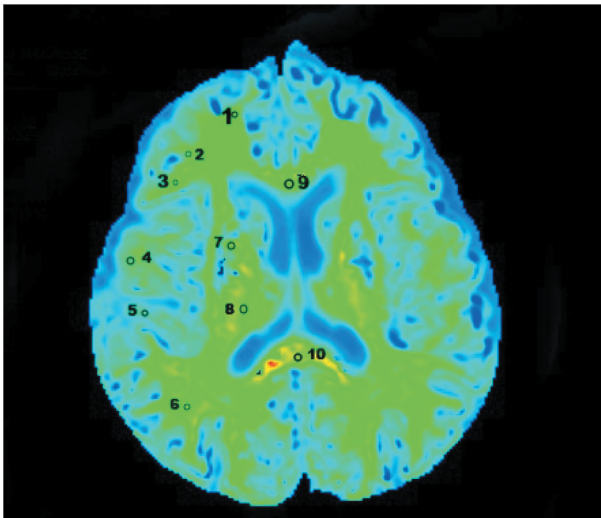


Figure 1. The placement of ROI: dot 1–10 represent superior frontal gyrus, middle frontal gyrus, inferior frontal gyrus, parietal lobe, temporal lobe, occipital lobe, the anterior and posterior limb of internal capsule, the genu and splenium of corpus callosum respectively.

were blind to the group distribution of research subjects.

2.4. Statistics

The statistical analyses were performed with SPSS 22.0. The quantitative data are expressed as mean $\pm$ standard deviation. One-way analysis of variance (ANOVA) was used to compare quantitative differences among three groups. For the factors that meet the homogeneity of variance data, the LSD method was used for comparison between groups; for those data that did not meet the homogeneity of variance, the Dunnett T3 test was used to compare the groups. Chi-square test was used to compare the differences between the groups in the classification data. The DKI date of each ROI and HAMD score was analyzed by Pearson or Spearman correlation. $p < 0.05$ was considered statistically significant.

3. Results

3.1. Demographic characteristics of subjects

The demographic data among these groups are comparable. There was no significant difference in age,

gender, marriage status, educational level, family history of psychiatric disorders among control, non-PSD and PSD groups (Table 1).

3.2. Stroke characteristics of subjects

The location, side and numbers of stroke between Non-PSD and PSD group were no significant difference (Table 2).

3.3. The decrease of MK/RK in frontal lobe, temporal lobe and/or genu of corpus callosum results in PSD

To indicate the damage to white matter microstructure was involved in PSD, MK/RK/AK detected by DKI in different groups were compared.

One-way ANOVA demonstrated that the MK value in L-SF ($F = 8.565$, $p = 0.001$) was associated with PSD, as well as in R-SF ($F = 18.431$, $p = 0.000$), L-MF ($F = 37.830$, $p = 0.000$), R-MF ($F = 7.536$, $p = 0.001$), L-T ($F = 19.544$, $p = 0.000$), R-T ($F = 19.686$, $p = 0.000$), G-CC ($F = 25.098$, $p = 0.000$). And the post-test analysis revealed significant decrease in MK at PSD group compared with Non-PSD and control (Figure 2).

One-way ANOVA of RK at PSD group in L-SF ($F = 9.573$, $p = 0.000$), L-MF ($F = 18.431$, $p = 0.000$), R-MF ($F = 18.431$, $p = 0.000$), L-T ($F = 19.544$, $p = 0.000$), R-T ($F = 19.686$, $p = 0.000$) demonstrated decreases relative to control and Non-PSD group (Figure 3).

There was no significant difference among control, non-PSD and PSD groups in AK (Figure 4).

3.4. The decrease of MK/RK in frontal lobe and temporal lobe was associated with the severity of PSD

Pearson or Spearman correlation analysis between MK/RK/AK and HAMD was utilized to clarify the relationship between the damage to white matter microstructure and the severity of PSD. The MK of L-SF ($r = -0.489$, $p = 0.002$), R-SF ($r = -0.510$, $p = 0.001$), L-MF ($r = -0.717$, $p = 0.000$), R-MF ($r = -0.346$, $p = 0.033$), L-T ($r = -0.666$, $p = 0.000$), R-T ($r = -0.602$, $p = 0.002$, Figure 5), G-CC

Table 1. Demographic characteristics of subjects.

		control	Non-PSD	PSD	χ^2/F	p
gender	Male	9(45%)	11(52.38%)	10(58.82)	0.709	0.712
	Female	11(55%)	10(47.62%)	7(41.18%)		
age		64.9 ± 9.72	64.71 ± 10.42	64 ± 10.95	0.038	0.963
marriage	Unmarried	1(5%)	3(14.29%)	2(11.76%)	1.100	0.577
	Married	19(95%)	18(85.71%)	15(88.24%)		
educational level	>9 years	4(20%)	8(38.1%)	4(23.53%)	1.848	0.432
	≤ 9 years	16(80%)	13(61.9%)	13(76.47%)		
family history of psychiatric disorders	yes	20(100%)	20(95.2%)	16(94.1%)	1.381	0.746
	no	0(0%)	1(4.8%)	1(5.9%)		

Table 2. Stroke characteristics of subjects.

		Non-PSD	PSD	χ^2	p
Location	hemisphere	17(80.95%)	13(76.47%)	0.113	0.737
	subtentorial	4(19.05%)	4(23.53%)		
Side	left	11(52.38%)	8(47.06%)	0.644	0.725
	right	9(42.86%)	7(41.18%)		
	bilateral	1(4.76%)	2(11.76%)		
Numbers	1	15(71.43%)	12(70.59%)	0.003	0.955
	≥ 2	6(28.57%)	5(29.41%)		

($r = -0.668$, $p = 0.000$) were negatively correlated with HAMD scores. The RK of L-SF ($r = -0.340$, $p = 0.037$),

L-MF ($r = -0.490$, $p = 0.002$), R-MF ($r = -0.457$, $p = 0.004$), L-T ($r = -0.467$, $p = 0.003$), R-T ($r = -0.676$, $p = 0.000$, [Figure 5](#)) were negatively correlated with HAMD scores. But there was no significant correlation between AK and HAMD score.

4. Discussion

In our study, DKI was used to explore the changes in white matter microstructure between PSD, non-PSD and control group. DKI can provide the kurtosis

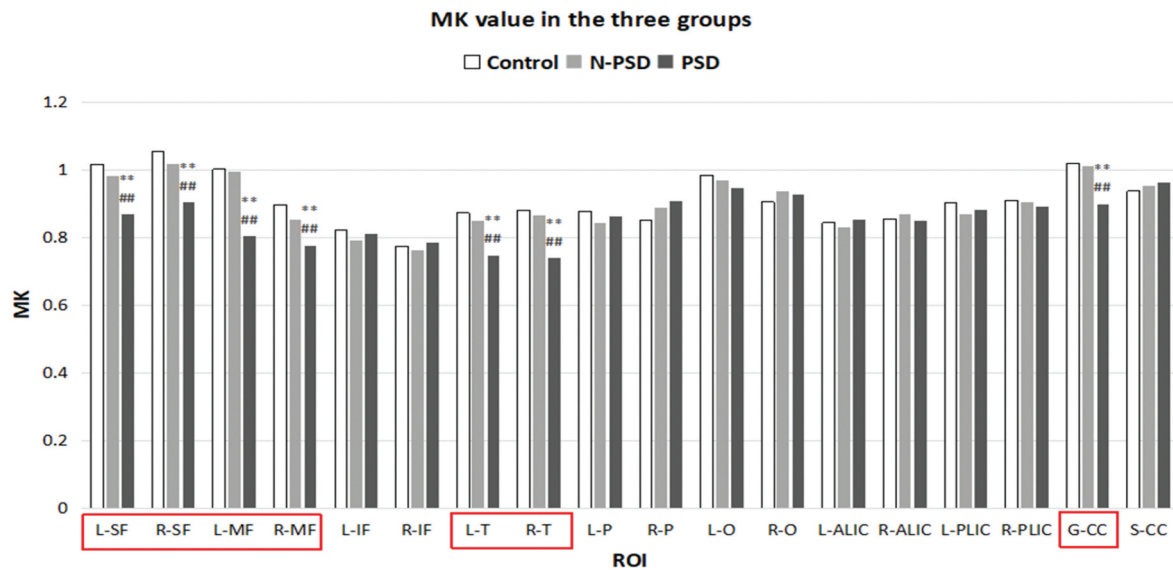


Figure 2. MK value in three groups: Compared with control and Non-PSD, MK of PSD group in bilateral superior frontal gyrus, middle frontal gyrus, temporal lobe and genu of corpus callosum was significantly decreased. *PSD vs control, #PSD vs Non-PSD, */# $p < 0.05$, **/## $p < 0.01$. **Abbreviations:** SF, superior frontal gyrus; MF, middle frontal gyrus; IF, inferior frontal gyrus; T, temporal lobe; P, parietal lobe; O, occipital lobe; ALIC, the anterior limb of internal capsule; PLIC, the posterior limb of internal capsule; G-CC, the genu of corpus callosum; S-CC, the splenium of corpus callosum; L, left; R, right.

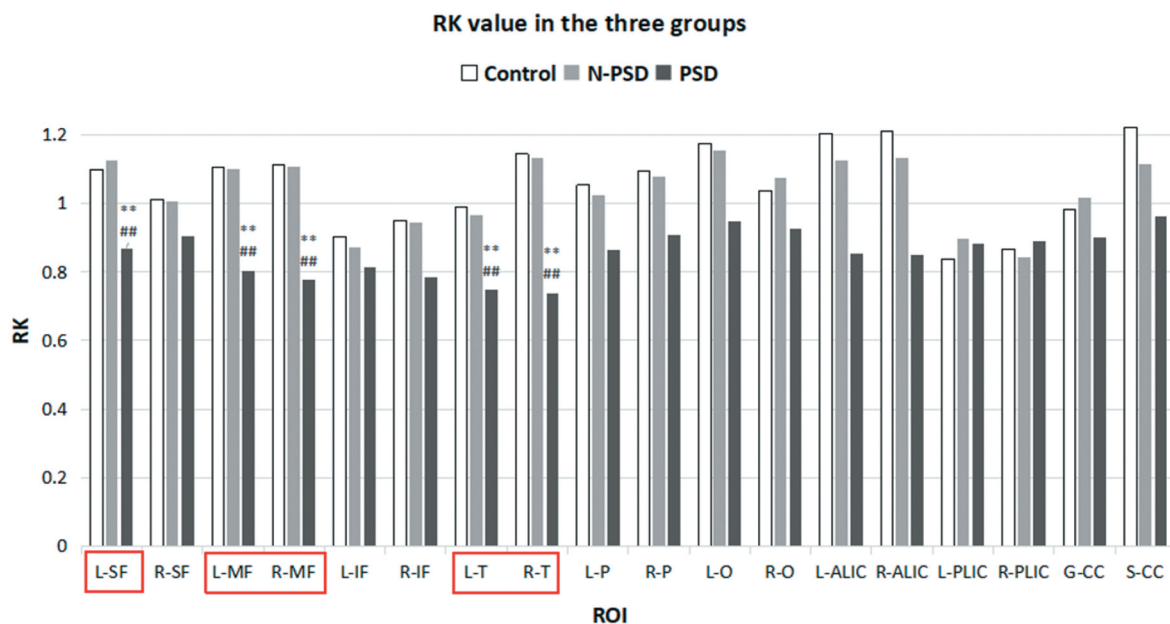


Figure 3. RK value in three groups: Compared with control and Non-PSD, RK of PSD group in left superior frontal lobe, bilateral middle frontal gyrus and temporal lobe was significantly decreased.

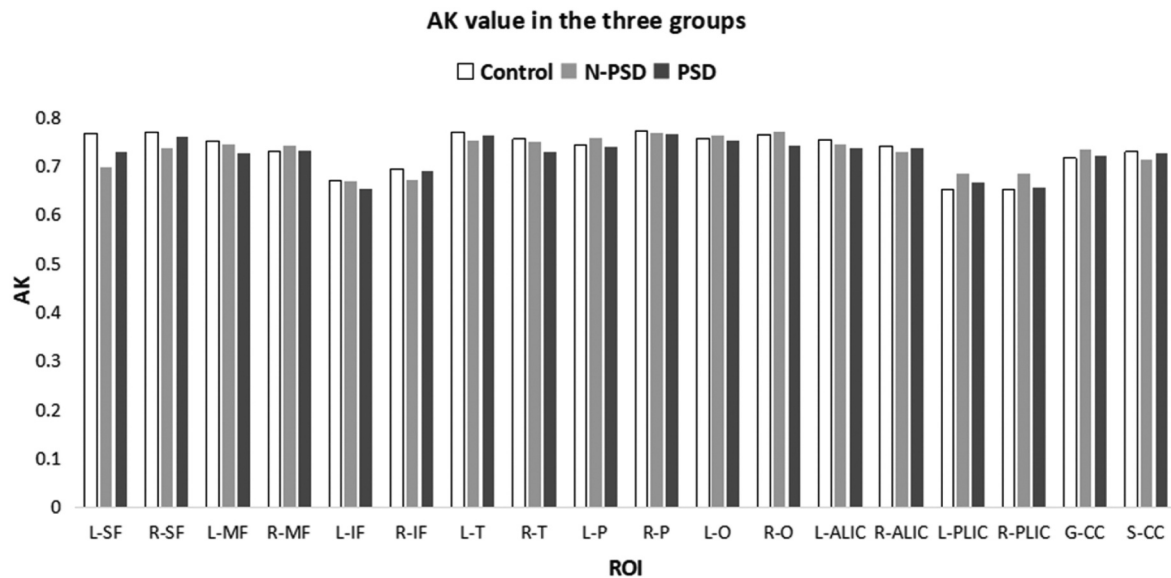


Figure 4. AK value in three groups: There was no significant difference among control, Non-PSD and PSD group.

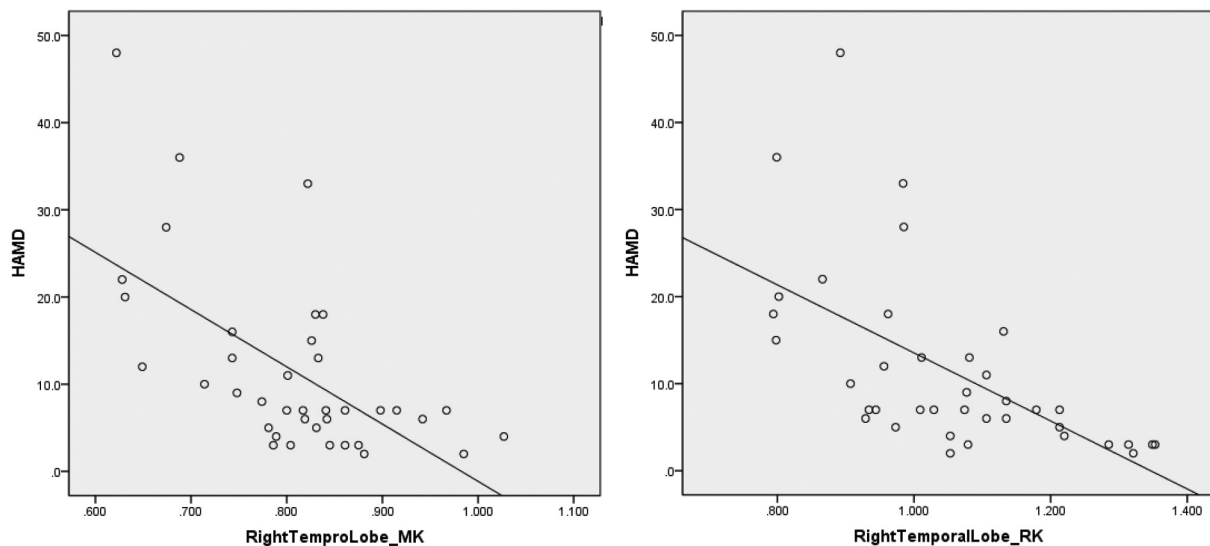


Figure 5. The correlation between MK/RK and HAMD: the MK (left)/RK (right) in right temporal lobe was negatively correlated with HAMD scores.(HAMD: Hamilton Depression Scale-24).

metrics: mean kurtosis (MK), radial kurtosis (RK), axial kurtosis (AK), fractional anisotropy of kurtosis (FAK) [22]. In this study, we selected three kurtosis parameters of DKI: MK, RK, AK, to evaluate the microstructure of white matter.

MK, the average diffuse kurtosis of the tissue along all direction, reflects the complexity of microstructure in ROI. MK is considered to be the most characteristic DKI parameter [23], which does not depend on the spatial orientation of organizational structure and can be used both in the gray and white matter[7]. The larger the MK is, the more restricted the diffusion of non-Gaussian distribution water molecules are, indicating the more complex of microstructure is [6]. Our results displayed that MK of PSD was decreased compared with control or non-PSD group (Figure 6) in

bilateral superior frontal gyrus, middle frontal gyrus, temporal lobe and genu of corpus callosum, suggesting that the occurrence of PSD may be due to the decrease in complexity caused by the destruction of white matter microstructure in frontal lobe, temporal lobe and genu of corpus callosum.

RK is the kurtosis information of diffusion which is perpendicular to axon direction. In white matter, RK is high because of the deviation of water diffusion from Gauss distribution as the result of the obstruction of cell membrane and myelin sheath [25]. In other words, the larger RK is, the better the integrity of cell membrane and myelin sheath are. We found that RK value of PSD group in left superior frontal lobe, bilateral middle frontal gyrus and temporal lobe was decreased. So we can speculate that the structure of

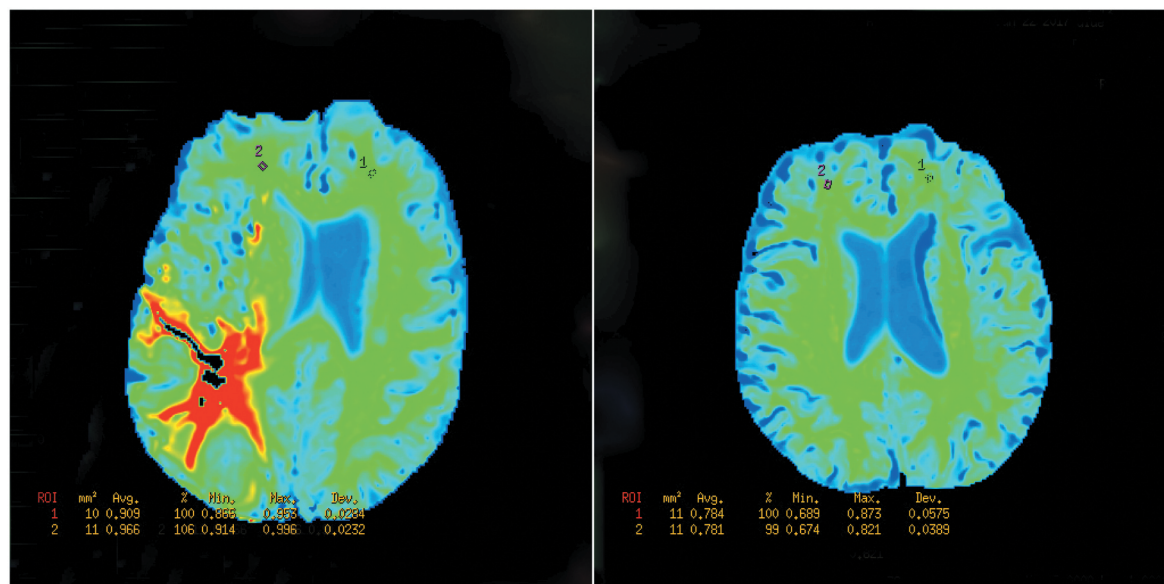


Figure 6. MK in Non-PSD (left) and PSD (right) group. Dot 1, 2 stand for the frontal lobe. Our results displayed that MK of PSD was decreased compared with Non-PSD group in bilateral frontal lobe ($P < 0.05$).

the cell membrane or myelin sheath in PSD patients is destroyed.

AK refers to the kurtosis information that diffuses along the axon direction. The restricted diffusion of water molecules in this direction is relatively few, so the AK is at a low level[3]). RK is believed to reflect myelin integrity while AK is typically believed to reflect axonal integrity[4]. In our study, there was no significant difference in AK between PSD and non-PSD group, is not consistent with other studies. Zhu J. J. et al. observed schizophrenia patients exhibited significantly decreased AK parameters in white matter regions, pointed out the decreased AK suggests axonal damage in schizophrenia patients [26]. But Delgado et al. found that there was nearly no contrast of AK parameter as a result of the overall absence of diffusion restriction in the axial direction [24]. Therefore, the role of AK in the evaluation of microstructure is needed for further study.

Our findings are similar to those of other researchers in patients with depression. Ota Miho et al. found MK values were significantly reduced in MDD patients in the right superior temporal cortex[6]. And depressive patients had lower values of MK and RK predominantly in the frontal lobe [21]. Therefore, we believe that the damage to white matter microstructure, predominantly in the frontal lobe and temporal lobe but also in genu of corpus callosum, is closely related to PSD.

Furthermore, we proved that MK in frontal lobe, temporal lobe and the genu of corpus callosum was negatively correlated with the severity of PSD by correlation analysis, so was RK in frontal lobe and temporal lobe. And in our previous studies, we have found that FA value of PSD patients in frontal lobe, temporal

lobe and genu of corpus callosum was decrease [27], therefore FA, MK, and RK can be combined to evaluate the occurrence and severity of PSD.

The change in white matter microstructure reflects the damage to cortical and subcortical LCSPT circuit in PSD patients. The LCSPT circuit is considered to be closely associated with emotional regulation, and PSD is a serious disorder of emotional regulation after stroke. Our study found that PSD group had lower MK, RK value than those in non-PSD or control group in frontal, temporal lobe and genu of corpus callosum.

As an important part of LCSPT circuit, frontal lobe has extensive connections with other cerebral cortex, limbic system and subcortical area [9]. DLPFC includes superior frontal gyrus and middle frontal gyrus, plays an important role in the regulation of emotion. Stroke in DLPFC may increase the risk of depression [10]. In our study, MK in bilateral superior frontal gyrus, middle frontal gyrus and RK in left superior frontal lobe, bilateral middle frontal gyrus was decreased. The decrease of MK/RK in superior frontal lobe and middle frontal gyrus suggested that fiber conduction pathway in DLPFC was damaged, which was closely related to PSD. Our study proved that frontal lobe, particularly DLPFC plays an important role in the occurrence of PSD.

The anterior part of the temporal lobe is the mental cortex [28]. Human emotional and mental activities are not only related to the prefrontal cortex, but also to the temporal lobe. The temporal lobe establishes extensive connections with other areas of the brain through the white matter fiber bundle to complete complex neural functions and participate in a variety of advanced neural activities, such as cognition, emotion, memory. The cortex of the temporal lobe is connected with the

cortex of the frontal, parietal and occipital lobes, and the hippocampus is connected with the basal ganglia and the limbic system. Consistent with that, we found MK and RK decreased in temporal lobe of PSD patients. Therefore, the occurrence of PSD may be related to the damage to temporal lobe.

Besides, the anterior and posterior limb of internal capsule and corpus callosum are important tissues connecting the bilateral neostriatum, thalamus and cerebral hemisphere. Corpus callosum is a transverse nerve fiber bundle and the largest commissural fiber in the hemisphere, which plays an important role in integration and transmission of information. White matter fibers communicate with the bilateral prefrontal cortex through the genu of corpus callosum. We found that MK value of PSD group in the genu of corpus callosum is lower, but there was no significant decrease of MK/RK value in anterior and posterior limb of internal capsule ($p > 0.05$), suggesting that the corpus callosum may be associated with the occurrence of PSD. However, there was no significant difference in FA value in corpus callosum between the depressive and control group in Nobuhara K's study [29]. Therefore, the role of corpus callosum and internal capsule in the development of PSD needs further study.

We utilized a new modality, DKI, to study the relationship between the change of white matter microstructure and PSD. DKI explores the structural changes of tissues from the micro-field, and its parameters, such as MK and RK, have better sensitivity and specificity than other imaging parameters, but DKI needs longer scanning time. In a word, DKI can be used for early diagnosis of PSD by detecting the changes of white matter microstructure. Our study also has some limitations such as: first, the small sample size, the lack of grouping comparison of severity of PSD and the absent of information on subtypes of ischemic stroke; second, the evaluation of depression two weeks after stroke is too short, we didn't have a long-term follow up; third, in the statistical method, there was no correction for multiple comparison; so further research is needed.

Conclusion

Our research indicated that the impairment of the white matter microstructure in the frontal lobe (particularly in dorsal lateral prefrontal cortex), temporal lobe and genu of corpus callosum may be involved in the occurrence of PSD, and the degree of damage is related to the severity of PSD. DKI may be an auxiliary diagnostic method for PSD by detecting the changes of white matter microstructure.

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Author contribution

All authors contributed to drafting and revising the article. All authors read and approved the final manuscript.

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